

MOLE CALCULATIONS-2

1. What is the mole number of $12,04 \cdot 10^{24}$ sodium atoms ?

- A) 20 B) 2 C) 0,5 D) 23 E) 200

2. How many atoms are there in 0,25 mole of silver ? (Avogadro's number, $N_A = 6 \cdot 10^{23}$)

- A) $1 \cdot 10^{23}$ B) $1,5 \cdot 10^{23}$ C) $3 \cdot 10^{23}$
D) $6 \cdot 10^{23}$ E) $24 \cdot 10^{23}$

3. What is the mass of 2.24 L of sulfur dioxide at STP?

- A) 3.2 g B) 6.4 g C) 9.6 g D) 16 g E) 32 g

4. Which one is the heaviest ?

- A) 3 moles of iron
B) 44,8 L of oxygen gas at STP
C) 2 moles of sulfuric acid
D) 150 g of calcium carbonate
E) 50 moles of hydrogen gas

5. Which one is **wrong** for 22 g of CO_2 gas ?

- A) It is 0,5 moles.
B) It occupies 11.2 L at STP.
C) It contains $3,01 \cdot 10^{23}$ of CO_2 molecules.
D) It contains $6,02 \cdot 10^{23}$ of oxygen atoms.
E) It contains 12 g of carbon atoms.

6. Which one of the gases given below has the greatest volume at STP ?

- A) 34 g H_2S B) 5 moles H_2 C) 8 g H_2
D) 22 g CO_2 E) 67.2 L N_2

7. 3 g of the compound X_2O has 1.4 g of X. What is the molecular weight of X_2O ?

- A) 18 B) 55 C) 30 D) 78 E) 94

8. What is the volume of C_2H_2 gas in liters at STP containing 0.25 mol of H atoms ?

- A) 22.4 B) 11.2 C) 5.6 D) 3.5 E) 2.8

9. Which one of the following substances given below has the greatest number of moles of atoms ?

- A) 1 mol Na B) 4.4 g CO_2 C) 0.2 mol CO
D) 0.1 mol O_3 E) 6.02×10^{23} molecules CO

10. I. An atom of nitrogen
II. One gram of nitrogen
III. A molecule of nitrogen

Find the correct increasing order of masses for nitrogen.

- A) I < II < III B) I < III < II C) II < III < I
D) III < II < I E) II < I < III

11. 1 mol of CO gas and 1 mol of CO_2 gas have the same at STP.

- A) mass B) number of atoms C) density
D) mass of O E) number of molecules

12. 2.24 L of XO gas has a mass of 3 g at STP. What is the atomic weight of X ?

- A) 14 B) 31 C) 1 D) 12 E) 32

13. What is the weight of one carbon atom if 7 mol of it weighs 84 g ?

- A) 1.99×10^{-23} B) 4.2×10^{-22} C) 6.02×10^{-23}
D) 4.2×10^{-23} E) 12

***14.** One molecule of a substance, X, has a mass of 7.89×10^{-23} what is the molecular weight of X?

- A) 47.5 B) 12 C) 44 D) 13 E) 38.8

***15.** What is the molecular weight of the compound, X_2Y , if mass of 3.01×10^{23} atoms of X is 0.5 g and 1.806×10^{24} atoms of Y is 48 g ?

- A) 12 B) 16 C) 24 D) 31 E) 18

***16.** 0.2 mol XO and 0.3 mol XO_2 has a total mass of 18.8 g. What is the atomic weight of X ?

- A) 24 B) 36 C) 12 D) 48 E) 32

***17.** Which one of the following substances contain Avogadro's number of atoms ?

- A) 1 mol H_2 B) 17 g NH_3 C) 20 g Ca
D) 0.5 mol O_2 E) 9 g H_2O

***18.** 22.4 L of a gas has a mass of 28 g at STP. What is the mass of one molecule of the gas ?

- A) 2.33×10^{-23} B) 4.65×10^{-23} C) 4.21×10^{-24}
D) 1.16×10^{-23} E) 1.69×10^{-25}